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PERFORMANCE SPECIFICATION

DETECTOR, MINE, METALLIC, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance requirements for the portable metallic mine detector.

2. APPLICABLE DOCUMENTS (Not applicable to this specification)

3. REQUIREMENTS

3.1 Description. The detecting set shall consist of a metallic-mine detector packaged in a transport case. The metallic mine detector shall consist of a search head, associated electronic components, a power supply, connecting cables, a collapsible handle and a headset. Throughout this military specification, the metallic mine detector will be referred to as the detector.

3.2 First article. When specified (see 6.2), a standard sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.3 System characteristics.

3.3.1 Performance.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications Electronics Command, ATTN: AMSEL-LC-LEO-E-EB, 10115 Gridley Road Ste 228, Fort Belvoir, VA 22060-5849 by using the Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3.1.1 Sensitivity. The detector shall detect the following targets described in table I, one at a time, in a plane parallel to the underside of the search head (see Appendix A, figure 1) and passing by the point(s) on the search head where a maximum signal is received at the following distances shown in table 1. Point(s) on the search head where a maximum signal is received shall be provided by the contractor.

TABLE I. Sensitivity targets.

TARGETS	DISTANCES
Aluminum hollow tube, Grade Alloy 5052 4.45 cm length, 0.64 cm dia., 0.177 mm + 0.05 mm thickness	13.5 cm
PMN - 6 striker pin 308 series (see figure 2)	7.5 cm
M-14 mild steel pin	5.0 cm

3.3.1.2 Detection capability.

3.3.1.2.1 Offset. The detector shall detect a simulated M-16 mine under the conditions of table II and at a minimum search head offset (from the edge of the search head to the center of the target) of not less than 15 cm.

3.3.1.2.2 No offset. The detector shall detect metallic objects as specified in the sensitivity requirements, (see 3.3.1.1) in sand, loam, magnetite soils at the contractor’s recommended specified sweep (see table II).

TABLE II. Target types and placement.

SOIL SWEEP SPEED TARGETS	SAND 50 cm/sec	LOAM 50 cm/sec	MAGNETITE 50 cm/sec
Aluminum, hollow tube	15 cm	15 cm	2.5 cm
PMN-6 striker pin, 308 series	10 cm	10 cm	2.5 cm
M-14 mild steel pin	surface	surface	surface
M-16 simulated mine	4 cm	4 cm	4 cm

The properties of the sand, loam and magnetite soils containing moisture from 2 to 25 percent and measured at frequencies of 300 MHz through 600 MHz are as follows:

Sand Relative dielectric constant of 2.57
Loam Relative dielectric constant of 3.79
Magnetite Magnetic susceptibility of 15k X 10^{-6} cgs

3.3.1.2.3 Immersed operation. When the search head receiver coil is submerged 1.5 meters below the air-water interface, the detector shall be able to detect metallic objects such as the M-14 mild steel pin.

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3.4 Environmental conditions.

3.4.1 Environmental operating conditions. The detector shall perform under the following conditions:

a. Soils of various types shall include clay, loam, sand and gravel, and magnetite (see 4.5.2 and table II).

b. Water immersion with the detector, including the headset, 1.5 meters below the air-water interface.

c. The detector shall be resistant to sand and dust. The sand and dust concentrations shall be 0.177 grams/cubic meter (natural conditions) at a velocity of 8 meters/second.

d. The detector shall operate between -32 to +51 °C.

e. The detector shall be able to withstand the humidity expected in the natural environment as described herein.

Temperature - °C	Humidity - %
26	95
30	85
35	74

3.4.2 Induced environment. The detecting set shall operate as required after exposure to the following induced environmental conditions:

a. The detecting set shall be resistant to vibrational stresses expected in shipment and use (see 4.7.2).

b. The detecting set shall withstand infrequent non-repetitive shocks or transit vibrations expected to be encountered in handling and service environments (see 4.7.3).

c. The detecting set shall not be damaged by bounce after tested in accordance with 4.7.2.3.

3.5. Operational characteristics.

3.5.1. Required operational characteristics. The detector shall be capable of the following characteristics when employed in a prescribed manner (see 6.4):

a. Having a sweep frequency over a 1.5 meter wide path of 1 complete sweep consisting of across the path, step, and back across the path per 6 seconds (maximum passover time).

b. Operation within 16 meters of another detector of the same model without experiencing or causing electrical interference between detectors.

c. Detecting but not activating a FFV-028 magnetic influence fuzed mine.

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d. Operating without visible or audible emissions which expose the user to enemy detection.

e. Providing not less than 16 hours of continual operation at an ambient temperature of 25 ±5 °C without degradation of the detector's performance before requiring battery replacement. A battery connection extension kit for body heat battery warming during cold weather operation or other provision may be supplied. Protection against battery installation with reversed polarity is required. Clear battery polarity markings are required on the batteries and battery casing.

f. Transmitting the detector output signal to the headset. The signal shall increase in intensity or frequency or both as the detecting head is moved closer to the metallic object.

g. Being operated for at least 20 continuous minutes without a degradation in performance due to aural or physical fatigue. The operator shall be able to resume mine detection without degradation in performance after a rest period of no more than 20 minutes.

h. Being easily operated from either the standing/walking or prone/crawling positions while wearing protective equipment Mission Oriented Protective Posture (MOPP IV) gear, cold weather gear, or helmets and flak jackets.

i. Detecting mines at the optimum performance when the search head is held at a height of 5 to 20 cm above ground surface and at a sweep speed of 50 cm/sec.

j. The detector shall be capable of being carried, assembled and operated by personnel wearing Nuclear, Biological, Chemical (NBC) protective equipment MOPP IV, cold weather gear, or helmets and flak jackets.

k. Having a greater than 92 percent probability of detecting standard metallic military mines and mines containing small metallic content buried in various types of soils as described in Section 3, Environmental Conditions, at depths consistent with doctrinal placement of mines when operated by a soldier, Military Occupational Specialty (MOS) immaterial.

3.6 Interface characteristics.

a. The detecting set, with its transport case, shall not exceed a total weight of 16 kilograms (35 pounds).

b. Controls shall be provided for:

(1) Power ON/OFF

(2) Headset loudness or sensitivity or equivalent automatic adjustment

(3) Control for making adjustments of operation for the detector shall be positioned such that the operator can perform any control adjustment in a standing and walking, or prone and crawling position. Controls and displays shall be clearly

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labeled to indicate the function served by the control or display. All labels shall be in English.

c. The detector shall be extendable such that the search head can be comfortably held at a sweep-height of 5 to 20 cm above the ground by a standing/walking user whose body is within the 5th percentile female to 95th percentile male. For use in the prone/crawling position, the detector shall be collapsible such that the distance between the base of the search head and the center of the user's hand shall be 0.5 meters or less when naturally gripping the detector.

d. The detecting set shall have a non-glare earth tone surface on all external parts (i.e. forest green, brown, black).

e. Each detecting set shall be permanently marked in one location with the manufacturer's CAGE code number, model number and serial number assigned by the manufacturer. Markings shall be in English.

f. The detector shall be powered by standard dry or alkaline batteries.

g. The detector shall not be damaged by any form of moisture.

h. The detecting set shall consist of the following components:

(1) Detector assembly with headset

(2) A transport case/shipping container that shall:

(a) be lightweight and re-usable

(b) be waterproof

(c) have latches that facilitate opening and closing while wearing Nuclear, Biological, Chemical protective equipment (MOPP IV) or cold weather gear.

3.7 Low voltage performance. The detector shall have a low-voltage aural indication. The detector shall exhibit no degradation of detection capability up to the point when the supply voltage decreases to a level that activates the low-voltage aural indication.

3.8 Maintainability. When tested in accordance with the Operational Tests (see 4.8) the Mean-Time-To-Repair (MTTR) of the mine detector at the unit level shall be 20 minutes or less with no special tools or equipment required.

3.9 Reliability. When tested in accordance with 4.14, the mine detector shall demonstrate a lower test Mean-Time-Between-Failure (MTBF) of 500 hours and an upper test MTBF of 1000 hours (see 6.6).

3.10 Safety.

3.10.1 Mechanical safety. Rotating and reciprocating components or components capable of exhibiting temperatures equal to or greater than 60 °C, exposed to contact by personnel, shall be guarded/insulated. Detector shall not contain toxic or radioactive materials.

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3.10.2 Electrical safety. The detector shall provide for protection of personnel from inadvertently contacting voltages capable of producing shock hazards, ensuring that all external parts, surfaces and shields, exclusive of antenna and transmission line are at ground potential at all times during normal operation. Plugs for use with metal cased detectors shall have provision for automatically grounding the metal frame or case of detector when the plug is mated with receptacle, and the grounding pin shall make first contact and break last.

3.10.3 Non-ionizing radiation safety. Detectors capable of emitting radio frequency radiation between 300 KHz to 100 GHz and having power density greater than or equal to l0 mW/cm2 shall not allow overexposure of personnel via design, fabrication, shielding or method of operating.

3.11 Interchangeability. All parts bearing the same part number shall be physically, electrically and functionally interchangeable.

3.12 Deterioration prevention and control. The detecting set shall be corrosion resistant or provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the detecting set may be exposed.

3.13 Desired interface characteristics. It is suggested that the detector meet the following characteristics:

a. field assembly time shall be 5 minutes or less while wearing NBC protective equipment (MOPP IV), cold weather gear, or helmets and flak jackets.

b. shall be operable and maintainable by 5th percentile female through 95th percentile male soldiers.

c. contain a pressure relief valve.

d. be hardened against NBC agents and the decontamination process and be decontaminable to negligible risk level.

e. chemical agent resistant paint.

3.14 Desired operational characteristics.

a. It is suggested that the detector have the ability to locate the center of a target by the x-y coordinate method (see Appendix A, figure 3).

b. It is suggested that the detecting set operate as required after exposure to a high altitude Ex-o atmospheric electromagnetic pulse.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

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4.2 First article inspection. When a first article sample is required (see 3.2), first article inspection shall be performed in accordance with table IV.

4.3 Conformance inspection. Conformance inspection shall include the examinations in accordance with table V.

4.4 Examination. Each detector shall be examined for compliance with the requirements specified in section 3 (see table III).

4.4.1 Sampling. The sample size of examination shall be determined from table III, column 2. The sample size for test shall be determined by table III, column 3. A sample shall consist of one or more mine detectors drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

TABLE III. Sample size.

Column I	Column II	Column III
Lot or batch size	Examination Sample Size	Test Sample Size
2 to 8	2	2
9 to 15	3	2
16 to 25	5	3
26 to 50	8	5
51 to 90	13	5
91 to 150	20	8
151 to 280	32	13
281 to 500	50	13
501 to 1200	80	20
1201 to 3200	125	32

TABLE IV. Verification and inspection - First article.

REQUIREMENT	**REQUIREMENT PARA**	**FIRST ARTICLE**	
		TEST PARA	**EXAM PARA**
Performance	3.3.1	4.5	
Sensitivity	3.3.1.1	4.5.1	
Detection	3.3.1.2	4.5.2	
No offset	3.3.1.2.2	4.5.3	
Offset	3.3.1.2.1	4.5.4	
Immersed operations	3.3.1.2.3	4.6	
Soils	3.4.1.a	4.5.2	
Water immersion	3.4.1.b	4.7.1.1	
Sand and Dust	3.4.1.c	4.7.1.2	
Temperature	3.4.1.d	4.7.1.3	
Humidity	3.4.1.e	4.7.1.4	
Vibration	3.4.2.a	4.7.2.1	
Shock	3.4.2.b	4.7.2.2	
Bounce	3.4.2.c	4.7.2.3	
Sweep frequency	3.5.1.a	4.5.2	
Proximity	3.5.1.b	4.11.1	

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REQUIREMENT	REQUIREMENT PARA	FIRST ARTICLE	
		TEST PARA	EXAM PARA
Magnetic fuzed mine	3.5.1.c	4.11.2	
Emissions	3.5.1.d	4.8.2	
Continuous operation	3.5.1.e	4.9	
Headset signal	3.5.1.f	4.8.2	
No human fatigue	3.5.1.g	4.8.2	
Ease of operation	3.5.1.h	4.8.2	
Optimum success rate	3.5.1.i	4.8.2	
NBC	3.5.1.j & 3.14.b	4.8.2 & 4.11.3	
Probability of detection	3.5.1.k	4.8.2	
Weight	3.6.a	4.12.1	
Controls	3.6.b		4.4
Extendable search head	3.6.c	4.8.2	4.4
Paint	3.6.d		4.4
Marking	3.6.e		4.12.2
Batteries	3.6.f		4.4
Components	3.6.h		4.4
Detector assembly & headset	3.6.h (1)		4.4
Carrying case lightweight	3.6.h (2)(a)		4.4
Carrying case waterproof	3.6.h (2)(b)	4.7.1.1	4.4
Carrying case latches	3.6.h (2)(c)		4.4
Low voltage performance	3.7	4.10	
Maintainability	3.8	4.13	
Reliability	3.9	4.14	
Mechanical safety	3.10.1		4.4
Electrical safety	3.10.2		4.4
Non-Ionizing safety	3.10.3		
Interchangeability	3.11		4.4
Deterioration prevention and control	3.12		4.16
Field assembly time	3.13 a	4.8.2	
Human factors	3.13.b	4.8.2	
Pressure relief valve	3.13.c		4.4
Desired operational - center target	3.14	4.15	

TABLE V. Verification and Inspection - Quality Conformance.

REQUIREMENT	REQUIREMENT PARA	QUALITY CONFORMANCE			
		Test			EXAM
		I*	S*	I*	S*
Performance	3.3.1				
Sensitivity	3.3.1.1	4.5.1			
Detection	3.3.1.2		4.5.2		
No offset	3.3.1.2.1		4.5.3		
Offset	3.3.1.2.2		4.5.4		
Immersed operations	3.3.1.2.3				
Soils	3.4.1.a				
Water immersion	3.4.1.b		4.7.1.1		

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REQUIREMENT	REQUIREMENT PARA	QUALITY CONFORMANCE			
		Test			EXAM
		I*	S*	I*	S*
Sand and Dust	3.4.1.c				
Temperature	3.4.1.d				
Humidity	3.4.1.e				
Vibration	3.4.2.a		4.7.2.1		
Shock	3.4.2.b		4.7.2.2		
Bounce	3.4.2.c				
Sweep frequency	3.5.1.a				
Proximity	3.5.1.b				
Magnetic fuzed mine	3.5.1.c				
Emissions	3.5.1.d				
Continuous operation	3.5.1.e				
Headset signal	3.5.1.f				
No human fatigue	3.5.1.g				
East of operation	3.5.1.h				
Optimum success rate	3.5.1.i				
NBC	3.5.1.j & 3.14.b				
Probability of detection	3.5.1.k		4.8.2		
Weight	3.6.a	4.12.1			
Controls	3.6.b				4.4
Extendable search head	3.6.c				4.4
Paint	3.6.d				4.4
Marking	3.6.e				4.4
Batteries	3.6.f				4.4
Marking	3.6.e				
Components	3.6.h				
Detector assembly & headset	3.6.h(1)				
Carrying case lightweight	3.6.h(2)(a)				
Carrying case waterproof	3.6.h(2)(b)				
Carrying case latches	3.6.h(2)(c)				
Low voltage performance	3.7		4.10		
Maintainability	3.8				
Reliability	3.9				
Mechanical safety	3.10.1				4.4
Electrical safety	3.10.2				4.4
Non-Ionizing safety	3.10.3				4.4
Interchangeability	3.11				
Deterioration prevention and control	3.12				
Field assembly time	3.13.a				
Human factors	3.13.b				
Pressure relief valve	3.13.c				
Desired operational - center target	3.14				

* I - Individual detector
* S - Sample in accordance with table III.

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4.5 Performance tests. Prior to starting the performance tests, the contractor shall describe in detail that information which constitutes a positive detection as heard in the headset (i.e., in amplitude, frequency, or other signal) while also specifying the point(s) on the search head where a maximum signal is received. Information shall also be provided on the frequency, amplitude, or other signal when the detector is in the saturated condition. The contractor shall provide information on the recommended head height for best detection results.

4.5.1 Sensitivity test. The test shall be conducted with a fresh set of batteries or suitable power supply. The headset output (i.e., amplitude, frequency, etc.) shall be measured and recorded by electronic instruments and compared to the information provided by the contractor prior to the start of the test.

a. Each detector shall be placed in a test jig fixture, at least 1 meter from metallic objects (excluding those which are part of the detector itself) and other extraneous magnetic/electrostatic fields.

b. The detector shall be powered up using the on-board batteries or suitable power supply and adjusted to the standard operational null condition.

c. The detector shall detect the targets described in table I, one at a time, in a plane parallel to the underside of the search head and passing by the point(s) on the search head where a maximum signal is received at distances shown in table I. Point(s) on the search head where a maximum signal is received shall be provided by the contractor. Less than 100 percent detection constitutes failure of this test.

4.5.2 Detection capability. Balance the detector with the search head not less than 1 meter from metallic objects. The detector shall be operational when subjected to the sand, loam, and magnetite test lanes in a controlled environment. Optimal head height shall be provided by the contractor. The detector shall detect the targets in table II when buried up to, and including, the maximum depths shown. Three of each type of target in table II shall be buried in each soil type for a total of 12 targets per soil type. This shall be referred to as a "lane". The detector shall detect the targets of table II when the search head receiver coil(s) is at a head height recommended by the contractor. The test shall be considered failed if, in any lane, any target type is not detected a total of seven or more times.

4.5.3 No offset. The detector shall detect the metallic objects listed on table II according to the specified depths, soils, and sweep speed. The detector shall detect the targets of table II when the search head receiver coil(s) is at a head height recommended by the contractor. A lane consists of three of each type of target in table II buried in each soil type for a total of 12 targets per soil type. A total of 37 passes shall be made over each lane. The properties of the sand, loam, and magnetite soils containing moisture from 2 to 25 percent and measured at frequencies of 300 through 600 MHz are as follows:

Sand	Relative dielectric constant of 2.57
Loam	Relative dielectric constant of 3.79
Magnetite	Magnetic susceptibility of 15k x 10^{-6} cgs.

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4.5.4 Offset capability. The detector shall detect a simulated M-16 mine. Failure to detect the M-16 simulated mine, under the conditions of table II shall constitute failure of this test.

4.6 Immersed operation. The detector shall detect a M-14 mild steel pin when the search head receiver coil is submerged 1.5 m below the water/air interface and is 7.6 cm above the test pin. The test shall be performed in a 3.5 percent synthetic sea-salt solution with a salinity pH value of 8 ±0.2 and at a temperature of 18 ±10 °C. After a 30-day waiting period, the detector shall be examined for corrosion. Evidence of corrosion that would affect the ability of the detector to perform its mission or failure to meet the criteria described in the sensitivity tests shall be cause for discontinuance of consideration of the detector. Any leakage or damage to the detector or failure to detect the M-14 mild steel pin shall constitute failure of this test.

4.7 Environmental conditions.

4.7.1 Operational conditions.

4.7.1.1 Leak test. The detector, with headset, shall be completely immersed in water so that all parts are 1.5 meters below the air-water interface for a period of two hours. The headset shall be removed after 5 minutes of immersion and allowed to dry for the remainder of the 2-hour immersion test. The temperature of the water shall be 18 ±10 °C. Bubbles emanating from within the detector shall be considered a leakage. However, bubbles which result from trapped air on the various exterior surfaces of the detector shall not be considered a leak. If leakage is observed, the test shall be discontinued. After the 2-hour immersion, the detector shall be assembled and tested in accordance with 4.5.1.a, b, and c, for all targets. Any leakage damage to the detector or failure to operate in accordance with 4.5.1 shall constitute failure of this test.

4.7.1.2 Sand and dust. The sand and dust concentrations shall be 0.177 grams/cubic meter (natural conditions) and tested with a velocity of 8 meters/second. The detector head and electronic package faces shall be oriented toward the blowing sand and dust for a minimum of 90 minutes. After the test, the detector shall be assembled and tested in accordance with 4.5.1 a, b, and c, for all targets. Heavy surface erosion/abrasion, penetration of seals, function impairment or failure to operate in accordance with 4.5.1 shall constitute failure of this test.

4.7.1.3 High/low temperature operation and thermal shock.

4.7.1.3.1 High temperature operation. Place the disassembled detector in the chamber and increase the temperature to +51 ±2 °C. Maintain this temperature for 3 hours, assemble, and remove the detecting set from the chamber. Test in accordance with 4.5.1, starting within 5 minutes and completing within 30 minutes after removal from the chamber. Return the detector to the chamber and duplicate this test for six continuous cycles. After the sixth cycle, test in accordance with 4.5.1.a through 4.5.1.c, using all targets. These tests shall start within 5 minutes after removal of the detector from the chamber. Failure to operate in accordance with 4.5.1 or

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inability to assemble the detector shall constitute failure of the high temperature operation test.

4.7.1.3.2 Low temperature operation. Place the disassembled detector in the temperature chamber and reduce the temperature to -32 ±2 °C. Maintain this temperature for 3 hours, assemble, and remove the detector from the chamber. Test in accordance with 4.5.1 using only the PMN-6 striker pin, 308 series (see figure 2), starting within 5 minutes and completing within 30 minutes after removal from the chamber. Return the detector to the chamber and duplicate this test for six continuous cycles. After the sixth cycle, test in accordance with 4.5.1 using all targets. These tests shall start within 5 minutes after removal of the detector from the chamber. Failure to operate in accordance with 4.5.1, or inability to assemble the detector shall constitute failure of the low temperature operation test.

4.7.1.4 Humidity. The detector, out of its transport case, shall be subjected to the following humidities along with their corresponding temperatures. This constitutes one cycle. This cycle shall be repeated 5 continuous times and tested in accordance with 4.5.1 using all targets specified herein after each complete cycle. After each cycle, the detector shall be assembled within 5 minutes after removing it from the chamber and tested in accordance with 4.5.1.a through 4.5.1.c, for all targets. Failure to operate in accordance with 4.5.1, after each cycle shall constitute failure this test.

Humidity - %	Temperature - °C	Time - Hours
95	26	6
85	30	2
74	35	8
85	30	2
95	26	6

4.7.2 Induced environments.

4.7.2.1 Vibration. The detector, out of its transport case, shall be mounted to the vibration table. It shall be vibrated along each of three orthogonal axes for a period of 1 hour, through a frequency range of 10 to 55 Hz, with a 0.25 cm double amplitude or maximum acceleration of 1.0g with a sweep time of 15.05 minutes logarithmically. The sweep time includes an ascending plus a descending sweep of the frequency range. After this test the detector shall be assembled and tested in accordance with 4.5.1.a, b, and c, for all targets. Any physical damage or failure of the detector to operate in accordance with 4.5.1 shall constitute failure of this test.

4.7.2.2 Shock. The detecting set shall be loaded, complete with accessories, in the transport case. A total of 26 drops, from a height of 122 cm, on all edges, corners, and faces shall be made onto plywood with a concrete base from a quick release hook or drop tester. The case shall be so oriented that, upon impact, a line from the struck corner or edge to the center of gravity of the case and its contents is perpendicular to the impact surface. After this test, the detector shall be assembled and tested in accordance with 4.5.1.a, b, and c, for all targets. Evidence of damage to any detector components or failure to operate in accordance with 4.5.1 shall constitute failure of this test. Damage to the transport case shall be allowed provided that it does not prohibit continuing its mission.

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4.7.2.3 Bounce. The detecting set shall be loaded, complete with accessories, in the transport case. Test the detecting set on a package-testing machine, with a plywood base, capable of maintaining a constant speed of 260 rpm (4-1/3 cycles per second) with an amplitude of 1.27 cm for a 2.54 cm excursion. Place the detecting set inside the "fence" with 2.54 cm spare or "throw". Bounce the detecting set in the three mutually perpendicular directions; that is, the ends, the sides, and the top and bottom of the detecting set case shall be subjected to the bumping, each side for a period of 15 minutes (total time 45 minutes). After this test, the detector shall be assembled and tested in accordance with 4.5.1.a, b, and c, for all targets. Any physical damage or failure of the detector to operate in accordance with 4.5.1 shall constitute failure of this test.

4.8 Operational characteristics.

4.8.1 Operational tests. User testing shall assess the ability of the soldiers, regardless of MOS, to operate, support and maintain each of the detector candidates in an operational environment during daytime and night time. Operator and maintenance training shall be conducted in accordance with the training plan for each type of mine detector. Mine detectors shall be employed in a manner as prescribed by the operating manual (see 6.4).

4.8.2 Operational detection capabilities. The detector shall have a greater than 92 percent probability of detecting standard metallic military mines and mines containing small metallic content buried in various types of soils at depths consistent with doctrinal placement of mines when operated by an MOS-immaterial soldier. The distance from the ground to the bottom of the head when the detector is employed shall be approximately 5 to 20 cm. Mine lanes shall require the mine detector to be used in areas of long grass, underbrush, leaves and stones, and with the detector head submerged in water.

The test shall evaluate the ability of soldiers to use the mine detectors when walking, standing, or in the prone/crawling position while wearing cold weather gear as well as MOPP IV gear. Testing shall be conducted in existing weather conditions.

During testing, maintenance and system support package adequacy data shall be collected and assessed. Machine/operator interface shall be of special interest during testing. Of particular concern is: the period of time an operator can continuously operate the mine detector before having to be replaced; the ease by which the operator is able to locate mines; the number of times the operator must adjust the null; and the ease of transporting, configuring, operating, and maintaining the detector.

4.9 Continual operation. Two detectors shall be provided with a fresh set of batteries and balanced, if required, in ambient conditions. The detectors shall be placed in saturation with a target placed at the point(s) on the search head where a maximum signal is received. Leave the detectors as such for 16 hours. At the 15-hour 45-minute mark, null the detectors, without changing batteries, and test in accordance with 4.5.1.a through 4.5.1.c. Failure to operate in accordance with 4.5.1 shall constitute failure of this test.

4.10 Low voltage performance. Place the mine detector in an ambient temperature of +25 ±5 °C. Adjust the power source voltage to the detector's standard operating voltage dc, balance the detector in air and reduce the

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power supply voltage until the low voltage aural indication is actuated. Record the voltage. Then raise the source voltage just enough to stop the low voltage alarm. Test in accordance with 4.5.1. Failure to operate in accordance with 4.5.1 shall constitute failure of this test.

4.11 Required operational characteristics.

4.11.1 Proximity interference. Place two detectors 16 meters ±10 cm apart such that neither detector is within 1 meter of any metallic object. Turn on and balance each detector. Aural indications in the headset shall be no different than if only one detector was in operation. Repeat above test with one detector in the target detecting mode. Any aural indication in either detectors' headset other than normal presentations, as stated by the contractor, shall constitute failure of this test.

4.11.2 Magnetic influence mine. An instrumented FFV-028 magnetic influence fuzed mine shall be buried flush with the soil-air interface. A flash bulb shall be connected to the mine to indicate actuation of the fuze. The detector shall detect but not activate the mine at a sweep speed of 50 cm/sec and at a head height ranging from 5 cm to 20 cm. Failure to detect the instrumented FFV-028 mine or actuation of the fuze by the detector shall constitute failure of this test.

4.11.3 Electromagnetic pulse. The detecting set shall be exposed to a high altitude Ex-o atmospheric electromagnetic pulse of 50 kV/meter for 30 nanoseconds. After exposure, the detector shall be tested in accordance with the sensitivity tests described herein. Failure to meet the success criteria of the sensitivity tests shall be noted for future reference.

4.12 Interface.

4.12.1 Weight. Weigh the detector and transport case. A weight of more than 16 kg (35 pounds) constitutes failure of this test.

4.12.2 Marking. Inspect the detector and transport case for clear and legible marking. Markings should be in English. Nonconformance to 3.6.e shall constitute failure.

4.13 Maintainability. Maintenance operations required during the operational testing shall be used to determine conformance to maintainability requirements specified in section 3. All maintenance actions performed during any testing (including scheduled maintenance) shall be timed and recorded in order to determine conformance to maintainability requirements specified in section 3. For each maintenance action, a complete record of the maintenance action required, the date and time of the occurrence, the cause for the maintenance record action, and the time to repair the detector shall be recorded. Nonconformance to 3.8 shall constitute failure of this test.

4.14 Reliability. Three detectors shall accumulate a total of 1920 hours of operation as described below. Each detector tested shall operate at least one-half the average of all mine detectors tested. An external power supply may be used. The following cycle shall be repeated until the total number of operating hours are completed: The detectors shall be turned off and on once every twenty minutes until 9 hours of operation are accumulated. The power off period shall be at least 30 seconds and shall not exceed one minute. The detector shall then be subjected to 15 hours of continuous operation followed

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by a test as specified in 4.5.1. Additionally, the search head of one mine detector shall be moved through its entire range of extension and resecured a total of 720 times (the same detector/head is to be exercised for the duration of the test). All failures will be scored by Government personnel. When tested in accordance with 4.5.1, more than two failures, loss of the ability to adjust or secure the search head handle or failure of the on/off switch shall constitute failure of this test.

4.15 Center determination. The detector shall determine the center of a target by using the x-y coordinate method. For targets less than or equal to 7.5 cm in diameter, the detector shall determine the center of the target within ±7.5 cm. For targets greater than 7.5 cm, the detector shall determine the center of the target within ±15 cm. Figure 3 illustrates this method. This is a desired feature and, therefore, has no failure criteria. However, the ability to accomplish this test will be noted for future reference.

4.16 Deterioration prevention and corrosion control. After conducting all the verification tests specified herein, inspect the detector and transport case for any signs of deterioration or corrosion. Nonconformance to 3.12 shall constitute a rejection of the item.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the inventory Control Point's packaging activity within the Military Department or Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. This specification covers a detecting set intended for use over all types of terrain for the location of buried metallic antipersonnel and antitank mines.

The US Army certified mine lanes located at CECOM Research Development & Engineering Center, Night Vision Directorate will be made available to all potential suppliers on a first come, first served basis. Reservations will be requested by letter addressed to NIGHT VISION DIRECTORATE ATTN: AMSEL-RD-NV-CD, 10221 GRIDLEY ROAD STE 430, FORT BELVOIR, VA 22060-5849. Requests for reservations should be received not less than two weeks in advance of required use. Confirmation will be by phone or facsimile. Information required includes the number of people, how long the mine lanes will be needed, and, if available, security clearances. If there are no security clearances, special escorts will be provided. The mine lanes will contain all the specified targets (table II), at the specified depths, that will be used during the first article tests. Extreme care will be used during use to ensure that the positioned targets are not disturbed.

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Note: If the supplier is from a foreign country, they must have prior clearance through their respective embassy to Headquarters, Department of the Army, before access can be granted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Whether batteries are to be furnished (see 3.6.f).
c. Time frame required for submission of the first article models and number of detecting sets to be furnished. Standard samples required (see 6.3).
d. Packaging data.

6.3 Standard sample. Standard sample detecting set will include a transport case, initial set of batteries, technical/operators manual, maintenance/repair manual, and all other components as specified in 3.1. The standard sample will be representative of a production lot of the specific detector model; i.e., it will be identical in design, construction, and performance. The quantity of detecting sets to constitute the standard sample will be specified in the contract or order.

6.4 Operating manual. The mine detector will be employed as prescribed in FM 20-32, Mines, Countermine Operations. Requests for this manual will be referred to NIGHT VISION DIRECTORATE, ATTN: AMSEL-RD-NV-CD, 10221 GRIDLEY ROAD, STE 430, FORT BELVOIR VA 22060-5849.

6.5 Packaging information. All components of a complete detecting set will be placed in the transport case. The case will be closed and secured. Each detecting set and all of the components will be preserved in an appropriate manner to afford protection against deterioration and damage from shipment from the contractor to the initial destination.

6.6 Definitions.

6.6.1 Failure. For purposes of determining the MTBF, a failure is defined as any malfunction, degradation, or performance outside specified limits which the operator cannot remedy by adjustment within 5 minutes using tools or parts supplied with the detector; or a hazard to personnel or equipment.

6.7 Subject term (key word) listing.

Detection
Sensitivity

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR4
Navy - SH

Preparing activity:
Army - CR4

Project 6665-0707

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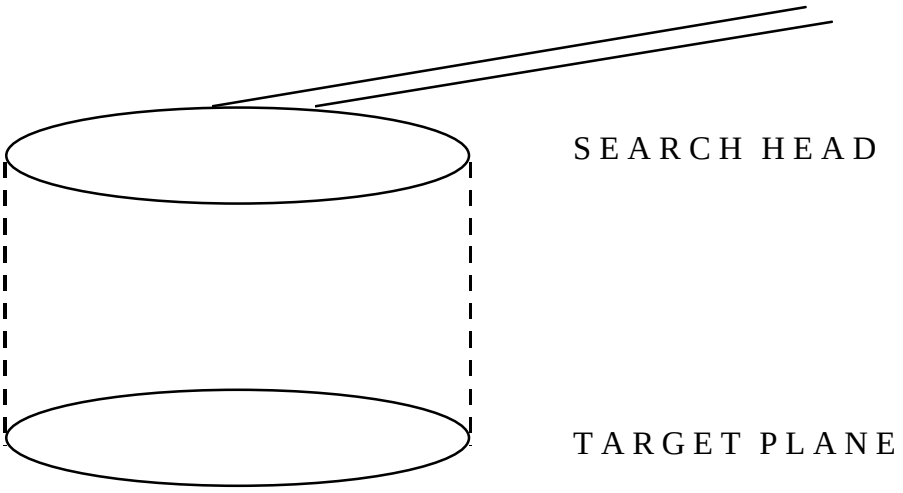


Figure 1. **Target Zone**

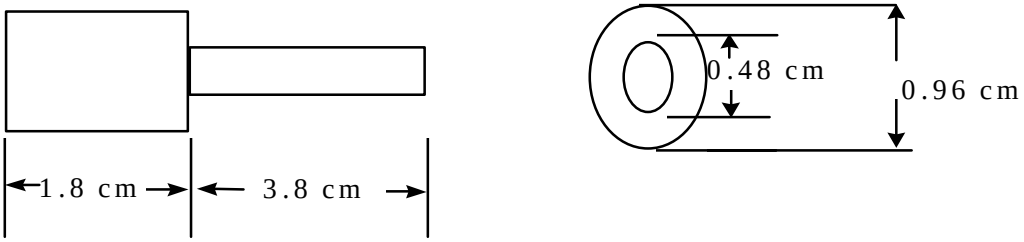


Figure 2. Striker pin

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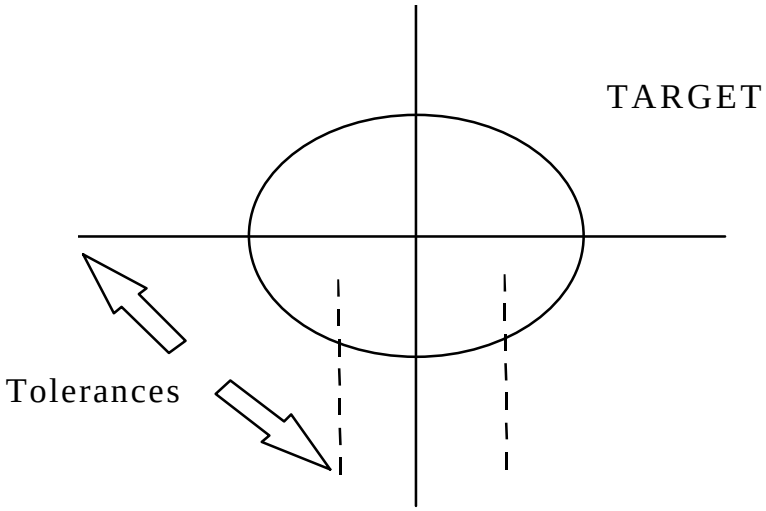


Figure 3. Center determination

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-23359H	**2. DOCUMENT DATE (YYMMDD)** 971119

3. DOCUMENT TITLE
Detector, Mine, Metallic, Portable

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME BELVOIR CECOM SOUTH	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) AUTOVON DSN 654-3468
c. ADDRESS (Include Zip Code) BELVOIR CECOM SOUTH 10115 GRIDLEY ROAD STE 228 ATTN: AMSEL-LC-IEW-D-ED Fort Belvoir VA 22060-5849	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, Oct 89 *Previous editions obsolete*

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